

Pascal Hafidz Fajri

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EDUCATION

University of Indonesia

Bachelor of Computer Science

Depok, Indonesia

2023 – Present

EXPERIENCE

Research Intern of MLCV Lab

Faculty of Computer Science, University of Indonesia

Feb 2026 – Present

Undergraduate Teaching Assistant

Faculty of Computer Science, University of Indonesia

Course: Programming Foundations 2

Feb – Jun 2025

- Explained and designed tasks around core programming concepts, including Object-Oriented Programming principles (encapsulation, inheritance, polymorphism, abstraction) using Java.

VOLUNTEER & ORGANIZATIONS

Software Engineering Staff

COMPFEST 17

Apr – Nov 2025

- Contributed to a large-scale full-stack event platform (React Router v7, NestJS, Prisma, PostgreSQL) spanning the Academy, Xcelerate, and Main Event program modules.
- Built the Academy student dashboard (notifications, profile, points-history) and its NestJS points-history API, powering the platform's rewards system.
- Developed the walk-in interview registration/queue flow and company exhibition pages for the Main Event program, backed by JWT-guarded RESTful NestJS APIs.

Community Service Department Staff

Student Executive Board – Faculty of Computer Science

University of Indonesia

Jun 2024 – Jan 2025

- PIC of “PEDAS” program — a fundraising initiative supporting the education of children of faculty staff (canteen workers, janitors, security personnel).

PROJECTS

ECG Waveform Digitization from Paper ECG Images | *Python, OpenCV*

Feb 2026 – Present

- Built an end-to-end pipeline to digitize paper ECG scans into aligned 12-lead time-series signals, covering preprocessing, baseline/separator detection, lead cropping, and calibrated voltage/time reconstruction.
- Improved robustness for noisy clinical scans through red-ink filtering, overlap detection, and iterative recovery for clipped or broken traces, then exported canonical NPZ outputs with interpolation and overlap masks for downstream ML tasks.

Vessel | *React, Next.js, Tailwind CSS, Rust, Solidity, Docker*

Dec 2025 – Mar 2026

- Contributed to a hybrid Web2/Web3 trade finance platform with “On-Chain Accounting”, bridging fiat (IDR) transactions with on-chain invoice tokenization (ERC-721) on Base Network.
- Built investor-facing funding flows and an admin KYC/business-verification dashboard on the Next.js/TypeScript frontend, plus Farcaster mini-app packaging.
- Implemented Rust backend fixes for invoice-approval error recovery and an on-chain shipment-verification call, and set up deployment (Docker, Fly.io).

VigilNet | *React, TypeScript, Capacitor, Fastify, Android*

2026

- Designed and implemented a BLE peer-to-peer crowd-density detection feature for an event-safety mobile app, building a native Android Capacitor plugin and its TypeScript bridge.
- Replaced a placeholder visualization on the organizer dashboard with a real Leaflet-based live map and density heatmap, and added a live alert-streaming backend endpoint.

Label Discovery on Indonesian Traditional Food Image Collection

Oct - Nov 2025

- Built a label discovery pipeline for the ACTION 2025 Data Mining competition, including preprocessing (outlier detection, smart cropping) and data augmentation to strengthen visual representations.

- Extracted features using pretrained vision models and applied over-clustering and ensemble clustering methods (KMeans, GMM, Agglomerative, Spectral) to uncover latent visual structures.
- Leveraged partial labeling to map over-clusters to 15 target classes, followed by semi-supervised refinement using Label Spreading for unlabeled data.
- Fine-tuned final image classifiers (ConvNeXt and other vision models) using refined labels, gaining 0.89+ accuracy.